A 57-cell container for every fixed 11-omino

A computer-assisted counterexample to the formula in OEIS A395495
Research candidate | 29 October 2026 | proof replay and final review completed

Abstract

We give a connected set of 57 lattice cells containing an integer translate of every edge-connected 11-cell set. Orientations are fixed and holes are permitted. This yields a(11) ≤ 57 for the translation-only universal-container problem. The formula a(n) = (n² - n + 6)/2 conjectured in OEIS A395495 predicts 58 at n = 11 and is therefore false. The proof reduces the assertion to 70 signed-diagonal domains: 35 empty-group cases, four forced-cell cases, and 31 finite potential certificates. We explain the certificate conditions and prove their soundness. The accompanying Lean sources close the literal endpoint without premises. No optimality of 57, first failing index, asymptotic statement, or algorithmic performance result is asserted.

1. Statement and witness

A cell is a point of Z^2, with adjacency at Manhattan distance one. A polyomino is a nonempty finite edge-connected set of cells. Let a(n) be the least size of a connected finite container C such that every n-cell polyomino P satisfies $P + t \subseteq C$ for some $t \in Z^2$. Each orientation is an input in its own right; placement permits translations only.

THEOREM. Put $D_5 = \{(x,y) \in Z^2 : |x| + |y| \le 5\}$ and
$C = D_5 \setminus \{(2,2), (2,-2), (-2,2), (-2,-2)\}$.
Then |C| = 57, C is edge-connected, and every 11-cell polyomino has an integer translate contained in C. In particular $a(11) \le 57 < 58$.

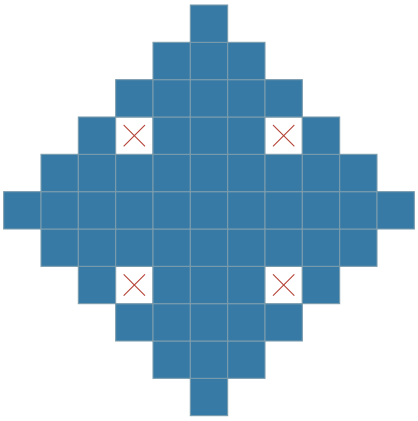


Figure 1. The 57 retained cells are blue; crossed squares are the four deleted cells of the 61-cell diamond. Cells denote lattice points, rather than an additional geometric restriction on their boundaries.

There are $1 + 4(1 + 2 + 3 + 4 + 5) = 61$ points in D_5. Connectivity survives deletion: decrease a nonzero absolute coordinate toward zero. If that step would enter a deleted point, the current absolute coordinates are (3,2) or (2,3); decrease the other coordinate instead. This remains in C and decreases |x| + |y|. Repetition reaches the origin. The containment assertion is proved in Sections 2-5.

2. Signed-diagonal normalization

Write u = x + y and v = x - y. Integer cells correspond exactly to pairs u ≡ v (mod 2); the inverse is x = (u + v)/2, y = (u - v)/2. An edge changes both signed coordinates by one in absolute value, and |x| + |y| = max(|u|, |v|).

For a connected 11-cell set P, let a and b be its u- and v-spans. Each lies between 1 and 10. Positivity follows from any edge. For the upper bound, choose cells realizing opposite extrema and take a simple path in P between them: it has at most ten edges, each changing either signed coordinate by one.

Reflection y ↦ -y swaps u and v, so we may arrange a ≥ b. If U = min u and V = min v, choose e ∈ {0,1} congruent to V - U modulo 2. Translation by (-U, e - V) in signed coordinates is integral in Cartesian coordinates because its two components have equal parity. The resulting set Q has extrema 0,a in u and e,e+b in v. Thus
Q ⊆ V(a,b,e) = {(u,v) : 0 ≤ u ≤ a,
e ≤ v ≤ e+b, u ≡ v (mod 2)}.
It meets each of the four extremal boundary lines.

Initially there are 2·(1 + 2 + ... + 10) = 110 triples with 1 ≤ b ≤ a ≤ 10. When e = 1 and a is odd, use
(u,v) ↦ (a-u, v-1).
Its Cartesian linear part is (x,y) ↦ (-y,-x), with integral translation ((a-1)/2,(a+1)/2). If instead b is odd, use
(u,v) ↦ (u, b+1-v).
Its linear part is (x,y) ↦ (y,x), with integral translation ((b+1)/2,-(b+1)/2). These maps preserve adjacency and normalize e to zero. Keep both parities only when a and b are even. The 55 span pairs plus 15 even/even pairs give 70 canonical domains. Coverage, rather than uniqueness of representatives, is sufficient.

RECOVERING THE ORIGINAL ORIENTATION. Every transformation above has the form Q = gP + d with g a square-lattice rotation or reflection and d an integer vector. The actual witness C is invariant under every such g: both the diamond and the four deleted points are invariant. If a normalized placement gives Q - c ⊆ C, apply g^(-1) to obtain
P + g^(-1)(d-c) ⊆ g^(-1)C = C.
The operation on the original piece is therefore a translation. This use of symmetry in the analysis does not authorize rotating a piece during placement.

The formal coverage layer uses literal Cartesian cells with signed-coordinate predicates. Its normalization theorems account for negative extrema and parity; a finite sample of normalized shapes is not a premise.

3. Every possible placement becomes a hitting constraint

For a normalized set with the four exact extrema of Section 2, a radius-five diamond center $c = (s,t)$ in signed coordinates must, and may, satisfy
$a-5 \le s \le 5$, $e+b-5 \le t \le e+5$, $s \equiv t \pmod 2$.
Let $S(a,b,e)$ be this finite set. These inequalities follow by applying the radius bound at both extrema of each coordinate. Conversely they place the whole domain inside the centered diamond. Thus no boundary center is omitted.

The deleted offsets become (4,0), (-4,0), (0,4), (0,-4). For each $c \in S$ form its collision group
$F_c = V(a,b,e) \cap \{c+(4,0), c-(4,0),$
$c+(0,4), c-(0,4)\}$.
Failure of every placement in C is equivalent to meeting every F_c. Add the four extremal-line groups. Any counterexample is now a connected 11-vertex set in the finite domain meeting all these groups.

Discarding duplicate groups, or a group containing a smaller retained group, preserves the hitting condition when every discarded group contains a retained one. More generally, to rule out a counterexample it suffices to use any selected subfamily of the original groups: hitting all original groups implies hitting this subfamily. The formal table instances prove membership in the literal original family individually. They do not assume that an external pruning algorithm is complete.

An empty collision group makes a counterexample impossible. This closes 35 domains. An empty CENTER SET is different: it leaves the four extremal groups. The domain (10,10,1) has no centers and is handled by a potential certificate, not discarded as a vacuous success.

4. The four forced-cell cases

Singleton groups force their vertices into any hitting set. In domain (7,3,0), sixteen distinct vertices are forced, contradicting size eleven immediately. In (7,4,0) and (8,3,0), ten vertices are forced. In (8,4,0), eight are forced. Exact finite exhaustion closes these three remaining cases: every completion to eleven vertices either is disconnected or misses an original group.

For orientation, the earlier independent mathematical checker exhausted supersets of size AT MOST eleven: respectively 0, 11, 9, and 576 supersets in the four domains, with 0, 0, 0, and 3 connected supersets. The three connected supersets in (8,4,0) miss a required group. It would be false to describe every superset there as disconnected. The formal proof instead exhausts exact eleven-cell completions using powerset-cardinality identities, certified cuts and missing-group checks; these are different exhaustion interfaces to the same obstruction.

The remaining 31 domains use Section 5. Thus 35 + 4 + 31 = 70 covers every normalized case. This arithmetic is a guide to the proof, not a substitute for the proved case disjunction.

5. Why the potential certificates prove impossibility

Work in a finite undirected graph on a domain D, with a nonempty finite set M of group labels and membership predicate H(i,z). For every nonempty $A \subseteq M$ and vertex $z \in D$, a table supplies a natural number L(A,z). The sufficient certificate conditions are:

BASE. If H(i,z) holds for every $i \in A$, then $L(A,z) \leq 1$.
EDGE. For every adjacent $z,w \in D$, $L(A,z) \leq L(A,w) + 1$.
MERGE. For disjoint nonempty $A,B \subseteq M$,
$L(A \cup B,z) + 1 \leq L(A,z) + L(B,z)$.
FULL. For every $z \in D$, $L(M,z) \geq 12$.

LEMMA. The first three conditions imply $L(A,z) \leq |T|$ whenever $T \subseteq D$ is nonempty and connected, $z \in T$, and T meets all groups in nonempty $A \subseteq M$. Consequently FULL excludes any connected hitting set of at most eleven vertices.

PROOF. Choose a spanning tree on T rooted at z. Assign each label $i \in A$ to one chosen witness vertex of T in its group. A vertex may receive many labels, but each label is assigned exactly once. For a vertex v let A_v be the labels assigned in its descendant subtree. Whenever A_v is nonempty, let K_v consist of v and all paths from v to its assigned witnesses. We prove $L(A_v,v) \leq |K_v|$ upward through the tree.

If v carries labels $B \neq \emptyset$, BASE supplies the initial bound $L(B,v) \leq 1$ on $\{v\}$. For every relevant child w, induction gives $L(A_w,w) \leq |K_w|$. EDGE gives $L(A_w,v) \leq |K_w| + 1$, a bound on $K_w \cup \{v\}$. If v has no assigned labels, initialize with the first relevant child; such a child exists because A_v is nonempty. Merge the remaining child contributions, and any initial labels at v, using MERGE. Their label masks are disjoint. Their vertex sets intersect exactly at v, so each subtraction of the common root counts the union correctly. This proves the inductive bound. A labeled leaf uses BASE alone; an unlabeled vertex with one relevant child uses EDGE alone. Unlabeled subtrees are ignored. At z we have $A_z = A$ and $K_z \subseteq T$, proving the lemma. Combining $L(M,z) \geq 12$ with $|T| \leq 11$ is impossible.

No rule for the empty mask, monotonicity in labels, disjointness of group memberships, or optimality of a computed recurrence is required. Disjoint ASSIGNMENTS of labels justify MERGE even when the groups overlap.

For the 31 finite instances, literal tables establish BASE, EDGE, MERGE and FULL. The generator is a data supplier; its search procedure is outside this inference. The earlier independent mathematical review checked 31 original tables totaling 1,874,144 byte entries, including all relevant base, edge and merge inequalities. The formal packet can select smaller original group subfamilies and recertifies their tables. The old entry count is not asserted as the size of the final formal table payload.

6. From finite data to closed Lean endpoints

The packet contains exactly 2022 consumed mathematical modules and the exhaustive FinalAudit module. All are preserved byte-for-byte from the root-frozen source manifest. Grid defines literal adjacency, finite edge-connectedness, translation and the universal-container minimum. Witness proves the stated cardinality and connectedness. The signed-normalization and coverage layers reduce arbitrary connected 11-cell sets to the 70 finite cases.

Potential proves a generic rooted-set construction from connectedness, then the potential bound. Its rooted constructors are singleton, one-edge extension, and union at a common root. This provides a formal counterpart of Section 5 without assuming a spanning tree as an endpoint premise. FiniteBridge transfers an injectively indexed finite domain, literal original groups, and the certificate into exclusion of an eleven-cell hitting set.

The packed layer proves encoding/decoding inverses, membership, union, intersection and complement identities, and table-row correspondence. Submask coverage and the symmetry of disjoint pairs justify checking only one ordering. Separate row bounds, BASE, actual-adjacency EDGE, MERGE and FULL checks are assembled using proved list take/drop identities. An external expected batch count is never a logical premise. Table literals and all checks are kernel checked; there is no native_decide or solver axiom.

The final namespace is
Polyomino57.Diagonal.FiniteProof.AllCases
with the following literal closed types (no additional hypotheses):
finiteCases : Polyomino57.Diagonal.FiniteCases
universalContainer : Polyomino57.UniversalContainer 11
Polyomino57.witness
eligible : Polyomino57.EligibleSize 11 57
minimum_le : Polyomino57.minimumSize 11 ≤ 57
source_refutation : ¬Polyomino57.SourceFormula
SourceFormula asserts the formula for all natural $n \geq 4$. An eligible 57-cell witness makes the least eligible size at most 57; substituting $n = 11$ into the formula gives 58, a contradiction. The default value used in the minimum definition when no eligible size exists is irrelevant because this witness establishes existence at eleven.

The audit enumerates actual module-origin kernel declarations, including private, generated and instance declarations. It prints all actual types, origins, transitive axiom closures and safety flags, and checks only propext, Classical.choice and Quot.sound. The inherited inventory has 5924 unique entries. There is exactly one partial flag exception: Polyomino57.attachCheck._unsafe_rec has unsafe=false, partial=true, axiom=false. The name is generated and does not mean the actual unsafe flag is true. All other entries must have all three flags false. No new exception is allowed.

The original manuscript worker replay, root's full extracted replay and independent final review replay have passed: 2022 mathematical modules plus FinalAudit and all 5924 actual records. Their receipts and audits are in replay-evidence/ and evidence/. All records have identical types, origins, axioms and flags. This status-only edition ran no new full replay; identity of all 2023 source hashes binds it to those completed replays. Root acceptance is tracked separately.

7. Source, chronology and limits of the claim

[1] Peter Exley, OEIS A395495, https://oeis.org/A395495 ; history: https://oeis.org/history?seq=A395495 . The inherited source investigation reports an entry authored 26 April 2026, read in full on 29 October before direct HTTP 403; the later refresh reconfirmed its current formula and conventions. The full all-n formula and translation-only convention are the source target. The entry discusses a holed n = 7 container; excluding holes would change its problem.

[2] MathOverflow question 420897 (2022), “What is the smallest size of a shape in which all fixed n-polyominos can fit?”, https://mathoverflow.net/questions/420897 . The inherited investigation read the question and three answer bodies. Those discuss the same optimization and asymptotic lower bounds, not a matching 57-cell construction at eleven. The listed small containers predate the 2026 OEIS entry. No priority over those constructions is claimed.

[3] Source-linked code: https://github.com/pexley-math/oeis-A395495 returned 404 in the inherited investigation. Its data and certificates were not compared. Companion A352029 concerns free pieces rather than this fixed-piece target; it is not a clearance source for the present result.

[4] “Smallest region of the plane that contains all free n-ominoes”, Code Golf (2018), https://codegolf.stackexchange.com/questions/167484 . The refresh read the construction and SAT-generation bodies: their orientations may vary within each free-piece obligation. The 37-cell eleven-piece container has only six rows and cannot contain the fixed vertical eleven-bar; the triangular construction explicitly transposes pieces. Neither is a fixed-piece subsumer.

[5] Gill Barequet, Solomon W. Golomb and David A. Klarner, “Polyominoes”, chapter 14 of Handbook of Discrete and Computational Geometry, third-edition preprint (2017), https://www.csun.edu/~ctoth/Handbook/chap14.pdf , Embedding Theorem 14.2.1 and Table 14.3.1. The refresh read the standard host bound $n^2 - n + 1$ (111 at eleven), not 57. Its printed coordinate cutoff has an indexing discrepancy, so no exact host-coordinate claim is inherited. These classical generation/optimization methods receive no new credit.

The 29 October source/prior refresh read the current OEIS entry and the closest bodies [2,4,5]. Complete named registry trees were refreshed at OpenAI 7ef5c0cc5ca4b0e072b7a050fca09ab24bc6fa3a, AIM 8729eab98bafe409b66c5e39253bf56569a62327, Count dbf673741a54c7dc9d88a6aec0e8148a538f4a3a, and Omega dev 66967414930e220080efa1c07df890cc81a521fb. The complete Omega D5/Problems trees replace the earlier bounded-diff residual at those heads. Exact URLs, dates, hashes and scope limits are preserved in evidence/. No matching construction was located. This is bounded no-match evidence, not a priority guarantee or a full-body sweep of every repository. Author code again returned 404, and fresh history access failed; frozen history observations do not close those gaps. A moving OEIS database footer is not evidence of an entry-specific mathematical change.

SUBSTANTIVE DISSENT. The refreshed independent reviewer recommends EXCLUDE for discovery credit: this is one small finite optimization correction to a recent extrapolation, with established search methods and punctured diamonds, rather than an optimum, scalable family or general principle. The reviewer found no mathematical defect or matching prior result. This dissent is preserved unaltered, not recast as unanimous eligibility clearance.

Root retained one-family eligibility subject to the final packet, with acceptance tracked separately: the full universal formula is refuted at an interior index; the classical tree-center bound gives 61, and no complete classical bridge to 57 was established. Standard enumeration and formalization receive no credit, but their use alone does not make the resulting counterexample a routine corollary. Root describes this as borderline and does not use proof size, effort or quota as evidence. No new algorithm or general structural result is claimed. A missed domain, wrong group meaning, failed symmetry recovery or dated same-object construction remains a decisive objection.

Proof replay and independent final review are completed. Novelty remains bounded by the documented searches and access gaps; the substantive dissent remains standing. Root owns acceptance, tracked separately. No optimality, least failure, general-n construction, asymptotic bound or performance claim follows.

8. Reproduction and resource qualifications

The accompanying README gives executable commands. Run replay.py with a new output directory. It asserts the SHA-256 and exact version of the supplied Lean binary, all eight dependency revisions in pins.json, the 2023 source-manifest entries and the 2022-module dependency plan. It creates and extracts a deterministic source-only replay archive, rejects candidate .olean/.ilean inputs, then compiles serially. Only pinned upstream dependency build caches are used. All candidate oleans are produced inside the new extraction.

Each Lean call uses -j1, -DwarningAsError=true and -DautoImplicit=false. Each stage has CPU 210 seconds, wall 230 seconds, and a process-group RSS kill threshold of 2800 MiB sampled every 0.045 seconds. Python helpers use 512 MiB address space with BLAS/OMP thread counts one. RSS sampling does not rule out an unsampled transient. The driver temporarily removes its own address-space restriction only across the launcher boundary so Lean is governed by the group RSS limit rather than the Python address-space limit; helper work remains bounded. It does not raise the Lean RSS/CPU/wall caps.

After all 2023 compilation stages, the parser verifies all 5924 distinct declaration records and actual flags/axioms, matching the expected name set, checks every mathematical log is empty, and rehashes the extracted formal sources. The final audit log retains the actual types and origins. Runtime reports and receipt are external replay evidence, not imported mathematical premises. A new extraction must be used for a fresh run.

The binary is Lean 4.34.1 (commit 5045d0056413266e57c625dcd7c365b10e377c52), SHA-256:
e8baaa71855a616dc351028f3ad2200051b
0671f423a1696a100e809302d5550
Mathlib is pinned at d13f23b723b8a846827a245b89c10fc7d3f11612. The other seven revisions are recorded in pins.json; every revision is checked by reading repository metadata, with no git operation.

The original manuscript input archive SHA-256 is:
31c9b601c4483faf2bed7739850eb401860
4c6fee2043af85355df22aac0bd90
The final packet manifest covers exact files, lengths and hashes. The PDF builder uses invariant ReportLab output and a vector cell diagram; the ZIP builder fixes entry order, timestamps, permissions and compression. Rebuilding from extraction must reproduce both bytes, with the relevant runtime/font versions held fixed and recorded in the build receipt.

Failures are evidence, not omitted runs. The inherited formal work exceeded the RSS limit on unsplit adjacency, BASE and forced-coverage checks; smaller proved slices resolved those failures without raising caps. It also corrected a rejected Cartesian/signed-center mismatch and proof-association error. The original manuscript lane's first version probe failed because Lean inherited the Python address-space limit; the corrected launcher isolates the policies. The original manuscript's PDF skill marker initially failed under Node's 512 MiB address space reservation, then succeeded under explicit CPU/wall and sampled 512 MiB RSS caps. No mathematical source was changed to address either packaging issue. Full inherited and local failure notes accompany the packet.

Root and independent final replays and the integrated final review are complete. The reviewer disclosed uncapped small metadata/text helpers; this prevents an unqualified all-protocol PASS. This edition adds no full replay and preserves those failures. Root's final packet inspection and acceptance decision remain separate.